SafeLane-RL: Constrained Reinforcement Learning for Highway Driving and On-Ramp Merging

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AI Course, Assignment 2, IIT Jodhpur GitHub: https://github.com/Osptium-Invincible/safelane-rl

Abstract—We cast tactical highway driving (speed control, lane changing, collision avoidance and on-ramp merging) as a constrained MDP and solve it with SafeLane: PPO with an ego-attention policy and a PID-Lagrangian safety constraint on time-to-collision. In highway-env we built an on-ramp scenario in which the ego must merge, fixed a neighbour-detection bug on merge roads, and compared 11 agents and 3 baselines over 26 scenarios (3 seeds, 49,250 test episodes). SafeLane had 0% collisions on the official highway-v0 with the highest return, and reached 97–99% merge success (PPO: 51%, gap-accepting rule-based driver: 88%). At twice the training traffic density it kept 94% success (rule-based: 72%).

I. Problem and RL formulation

Environment. We use highway-env 1.12.1: a kinematic bicycle model whose traffic follows IDM (car following) and MOBIL (lane changes). The policy acts at 1 Hz; physics runs at 5 Hz in training and 15 Hz in evaluation. The scenarios are Highway (train: 3 lanes, 20 cars; test: official highway-v0, 4 lanes, 50 cars, 40 s) and our RampMerge. In RampMerge the ego starts on the on-ramp and must merge within an 80 m zone before a barrier; success means it merged and passed x=370 m without crashing. Stock merge-v1 instead puts the ego on the main road. Its connected-lane neighbour search links road segments by lane index (the ramp is lane 0, then lane 2), so we link them geometrically. Training traffic is domain-randomised per episode: density in $[0.7, 2.0]$ (highway) or $[0.5, 1.5]$ (ramp), and a driver population drawn from {IDM, randomised, aggressive, cautious}. Each randomised car samples its own time gap (0.5–2.5 s), politeness and comfort acceleration.

State $\mathcal{S}$. A Kinematics matrix $s_t \in [-1, 1]^{V\times F}$: the ego and its $V-1$ nearest cars, relative to the ego and normalised with a fixed range on every road, with empty rows set to zero. MLP agents use V=5, F=5 (presence, x, y, v_x, v_y); attention agents use V=15, F=7 ($+\cos\psi, \sin\psi$). Only cars within 200 m are visible, so the problem is a POMDP. Action $\mathcal{A}$. Five discrete meta-actions {lane-left, idle, lane-right, faster, slower} over target speeds $\{10, \dots, 30\}$ m/s, tracked by PID controllers. Reward R. Highway (default): $r_t = \mathrm{lmap}_{[0,1]}(-\mathbb{1}_{crash} + 0.1\,\ell_t/(L-1) + 0.4\,\tilde{v}_t)\mathbb{1}_{road}$, where $\tilde{v}_t$ maps $[20, 30]$ m/s to $[0, 1]$. Ramp: $r_t = -\mathbb{1}_{crash} + 0.2\,\tilde{v}_t + \mathbb{1}_{merge}$. A crash terminates the episode, so it also forfeits all future reward. γ=0.9, an effective horizon of about 10 s.

Constraint. $\max_\pi J_r(\pi)$ s.t. $J_c = \mathbb{E}\sum_t c_t \le d$=2, with $c_t = \mathbb{1}[\mathrm{TTC}_t<2\,\mathrm{s}] + \mathbb{1}[h_t<1\,\mathrm{s}] + 10\,\mathbb{1}_{crash}$ (TTC $= \Delta x/\Delta v$, headway $h = \Delta x/v$). PPO maximises $(r_t - \lambda c_t)/(1+\lambda)$. After every 1,024-step rollout, $\lambda \leftarrow [K_p e_k + K_i \sum e]_+$ with $e_k = \hat{J}_c - d$ (PID-Lagrangian).

Agents.

- Value-based: DQN; our Double+Dueling 3-step DQN (D3QN); QR-DQN.
- PPO family: PPO with an MLP; PPO with our ego-attention extractor (the ego token queries all cars, 2 heads, empty slots masked, 46.7k parameters versus 146k for the MLP); PPO-Attn with a density curriculum; PPO-Attn with a fixed cost penalty (λ=0.1); MaskablePPO with a TTC action shield (ablation).
- Ours: SafeLane = attention + PPO-Lagrangian, and SafeLane-Mixed, trained on both tasks.
- Baselines: random, constant speed, and rule-based IDM+MOBIL; on the ramp it matches the target lane's speed and merges by critical-gap acceptance (0.8 s).

Budgets are 50k steps for the DQN family and 120k for the PPO family (8 envs), plus a "long" round at 400–500k steps. Every method uses 3 seeds. Total compute was 10.5 CPU-hours on a laptop; no GPU is needed, because the simulator dominates.

II. Experiments and results

Each agent is evaluated on 50 episodes per scenario with held-out seeds and deterministic actions. We report means and 95% hierarchical bootstrap CIs (resampling seeds, then episodes), plus IQM. The 26 scenarios cover:

- Traffic: densities 0.5–2.5 (highway) and 0.5–2.0 (ramp), including values outside the training range; aggressive and cautious drivers.
- Perception faults: sensor noise $\sigma \in \{0.05, 0.1, 0.2\}$, 30% vehicle dropout, a 1-step delay.
- Transfer: stock merge-v1.

Highway. SafeLane had no collisions in 150 highway-v0 episodes and the highest return (33.0). It kept headway below 1 s only 1% of the time, against 20–25% for PPO and PPO-Attn. Attention alone was not enough: PPO-Attn without the constraint crashed in 18% of episodes, with large seed variance. The rule-based driver was also crash-free but less efficient (return 30.1). Constraint vs. penalty vs. shield. Unconstrained PPO agents drive at about 28 m/s and crash in 22–49% of merges. A well-chosen fixed

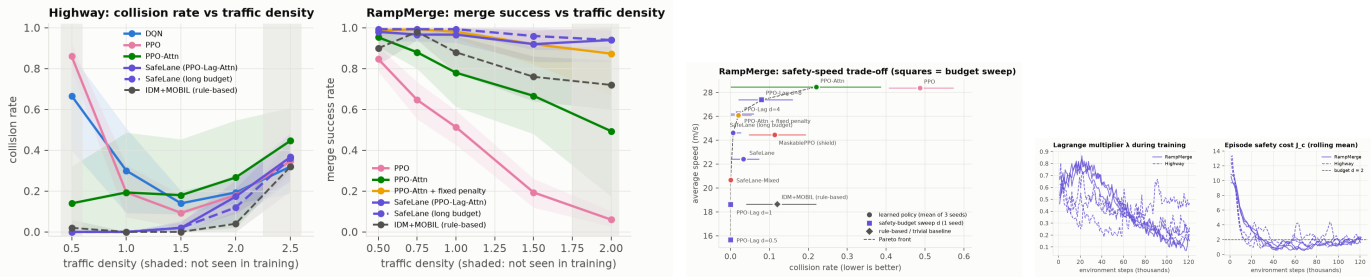


Fig. 1: Left: robustness to traffic density (shaded regions were not seen in training). Middle: speed versus collision trade-off on RampMerge; squares are SafeLane trained with budgets $d \in \{0.5, 1, 4, 8\}$. Right: Lagrange multiplier λ and episode cost during training; the cost is driven to the budget d=2 and λ settles near 0.5.

TABLE I: Main results (%, return). H: highway-v0; $H_{0.5}$: half density (outside training); R: RampMerge; $R_{2\times}$: double density (outside training). Mean of 3 seeds × 50 episodes.

	H coll.	H ret.	$H_{0.5}$ coll.	R succ.	R coll.	$R_{2\times}$ succ.
Random	72.0	12.9	42.0	30.0	70.0	20.0
Constant speed	96.0	14.2	18.0	0.0	100.0	0.0
IDM+MOBIL (rules)	0.0	30.1	2.0	88.0	12.0	72.0
DQN	8.7	30.1	66.7	–	–	–
D3QN (ours)	36.0	25.0	36.7	87.3	12.7	60.7
QR-DQN	14.7	29.2	68.0	–	–	–
PPO	33.3	28.0	86.0	51.3	48.7	6.0
PPO-Attn	18.0	29.7	14.0	78.0	22.0	49.3
PPO-Attn+penalty	–	–	–	98.0	2.0	87.3
MaskablePPO (shield)	–	–	–	88.7	11.3	70.0
SafeLane	0.0	33.0	0.0	96.7	3.3	94.0
SafeLane (long)	0.7	32.8	0.0	99.3	0.7	94.0
SafeLane-Mixed (long)	3.3	31.0	0.0	98.7	1.3	82.0

penalty matches SafeLane on success at normal density (98.0 vs 96.7%, P(SafeLane>penalty) = 0.38), but drives more aggressively (cost 3.2 vs 1.2, near-misses 19 vs 7%) and degrades more at 2× density (87 vs 94%). The learned λ settled at task- and seed-specific values (0.12–0.47) that no single hand-picked weight matches. The shield backfires in dense traffic: 43 of MaskablePPO's 45 crashes at 2× density are into the end-of-ramp barrier. It blocks unsafe merges, but the agent never learns to wait for a gap.

Generalisation. At half density (outside training), MLP agents crash in 67–86% of episodes by accelerating into a lone slower car, while SafeLane crashes in 0%. On the ramp at 2× density, SafeLane keeps 94% success, against 6% for PPO, 49% for PPO-Attn, 45% with a curriculum and 72% for the rule-based driver. SafeLane-Mixed solves both tasks with one policy (99% highway, 100% ramp). Specialists fail at the other task (0–67%). Budget as a dial. Varying d traces a Pareto front from 100% success at 15.7 m/s (d=0.5) to 27.4 m/s with 8% crashes (d=8). Long training raised SafeLane to 99.3% at 24.6 m/s, but reduced its robustness to sensor noise. Explainability. About 75% of the attention goes to the 3 nearest cars. Integrated Gradients show that decisions depend on the positions of the cars around the target gap.

III. Discussion and limitations

A constraint with a learned multiplier gave the most reliable safety across tasks and traffic levels, without tuning a penalty weight. Attention was essential when traffic was sparser or denser than in training. An internal review found that highway-env chose the observation scale per episode, and that the traffic speed factor was applied twice; we fixed both and retrained everything. Limitations.

- Every agent was trained on perfect observations and is fragile to sensor noise: SafeLane has 29% highway collisions at σ=0.1. The rule-based driver reads the true simulator state, so it is unaffected.
- The constraint holds on average, not per episode, and in very dense highway traffic the rule-based driver crashes less (4 vs 17% at 2×).
- The simulator is a 2-D world with simulated drivers.
- We used 3 seeds, and the budget sweep used 1 seed.

Future work: noise-randomised training, an MPC or CBF safety filter, real traffic data (highD), and multi-agent merging.